**MSDM 5004 Past Final Exam Paper 2026**

**1.** Consider the equation $u_t + au_x = 0$, where $a$ is a constant.

**(i) (15 points)** Show that this scheme is first order in $t$ and second order in $x$.

**(ii) (20 points)** Use Fourier analysis to find the stability condition for the scheme:

$$\frac{1}{\Delta t}(U_j^{n+1} - U_j^n) + \frac{a}{2\Delta x}(U_{j+1}^{n+1} - U_{j-1}^{n+1}) = 0.$$

**2. (15 points)** Calculate the results of the first two iterations using the Gauss-Seidel method in solving the following boundary value problem of the Poisson equation using the five-point scheme. Use a uniform square grid of size 1.

$$u_{xx} + u_{yy} = x + y, \quad 0 \le x \le 3, \ 0 \le y \le 3,$$

The boundary conditions are $u(0, y) = u(3, y) = u(x, 0) = u(x, 3) = 0$. Choose 0 value for the initial estimates in the iterations. Keep 4 significant digits in the results.

3. Consider the signal

$$h(t) = 2\cos(3\pi t)\cos(5\pi t).$$

The signal is sampled at $t = 0, 0.25, 0.5, 0.75$. The four sampled data are denoted by

$$h_0 = h(0), h_1 = h(0.25), h_2 = h(0.5), h_3 = h(0.75).$$

(a) Write down the values of $h_0, h_1, h_2, h_3$.

(b) What is the Nyquist critical frequency $f_c$?

The discrete Fourier transform (DFT) of $h_k$ ($k = 0, 1, 2, \ldots, N-1$) is given by

$$H_n = \sum_{k=0}^{N-1} h_k e^{2\pi ikn/N}.$$

(c) Obtain $H_0, H_1, H_2, H_3$ from the above formula. Also write down the corresponding frequencies of the DFT inside the range $-f_c < f \le f_c$.

(d) Account for the value of each of the $H_n$ obtained.
[Hint: Product-to-Sum formula of trigonometry and/or Euler's formula]

4. Consider the probability distribution function (PDF)

$$p(x) = Cxe^{-2x} \qquad x \in [0, 1)$$

(a) Show that

$$C = \frac{4e^2}{e^2 - 3}$$

(b) Explain why the simple transformation method cannot be applied to sample the above PDF.

Suppose one samples the distribution $p(x)$ by rejection method using the comparison function

$$f_1(x) = Ce^{-2x} \qquad x \in [0, 1)$$

(c) Show that $f_1(x)$ is a valid comparison function.

(d) State clearly the steps of the rejection method using this comparison function.

(e) Find the rate of acceptance.

Suppose one samples the distribution $p(x)$ by rejection method using the comparison function (which is not a good choice)

$$f_2(x) = Cx \qquad x \in [0, 1)$$

(f) Show that $f_2(x)$ is a valid comparison function.

(g) State clearly the steps of the rejection method using this comparison function.

(h) Find the rate of acceptance.

(i) State whether one can determine which of the above choices of comparison functions is better (I) based on the answers of (e) and (h) only, or (II) that one should take into account other factor(s) as well. If your answer is (I), state which choice is better with explanation. If your answer is (II), give an example of such additional factor(s).

5. (a) Show that

$$\int_{-h/2}^{h/2} f(x)dx = f(0)h + \frac{1}{24}f''(0)h^3 + O(h^5)$$

[Hint: Taylor series expansion]

(b) Prove the composite mid-point rule with leading error term:

$$\int_a^b f(x)dx = h\left[f_{1/2} + f_{3/2} + \cdots + f_{N-1/2}\right]$$

$$+\frac{f'(b) - f'(a)}{24}h^2 + O\left(h^4\right)$$

where

$$h = \frac{b-a}{N}$$

$$f_{i+1/2} = f\left(a + \left(i + \frac{1}{2}\right)h\right), \qquad i = 0, 1, 2, \cdots N-1$$

Consider the integral

$$I = \int_0^2 e^{-x^2} dx.$$

(c) Calculate $I$ using the composite mid-point rule with $N = 2$. Keep at least 6 sig. fig.

(d) Calculate $I$ using the composite mid-point rule with $N = 6$. You *must* use the answer of (c). Keep at least 6 sig. fig.

(e) Using the answers of (c) and (d), construct an answer which has error of $O(h^4)$ instead of $O(h^2)$. Keep at least 6 sig. fig.

Consider the multi-dimensional integral

$$I = \int_{-2}^{2}\int_{-2}^{2}\int_{-2}^{2}\int_{-2}^{2} e^{-x_1^2-x_2^2-x_3^2-x_4^2} f(x_1, x_2, x_3, x_4)dx_1 dx_2 dx_3 dx_4.$$

To evaluate the integral by Monte-Carlo integration, 10 points are sampled with distribution

$$w(x_1, x_2, x_3, x_4) \propto e^{-x_1^2-x_2^2-x_3^2-x_4^2}$$

inside the hypercube with side length 4 and centered at the origin.

The values of $f(x_1, x_2, x_3, x_4)$ are evaluated at the 10 points to yield:

0.1546, −0.1574, 0.2589, 0.9951, 0.3573, −0.2997, 0.8922, 1.8327, 0.2034, −0.5601

(f) Calculate the integral by Monte-Carlo integration with error estimation.